

SJAA Ephemeris Bulletin

✱ Sept 2025

PAH

Carbon, Hydrogen, and Oxygen make up 93% of the human body. As we learned from one of our most recent Astro 101 talks (Gravity, Rockets, Rogues and Aliens), these elements are among the most common in the universe. They exist in stars, in the interstellar medium (aka space dust), and in the gas that makes up some of the brightest deep sky objects that we love to observe. What, then, separates human beings from space dust?

Perhaps not as much as you may think. Scientists hypothesize that the same or similar hydrocarbons found throughout the galaxy were also key ingredients in the “primordial soup” from which organic life arose here on Earth. Specifically, **Polycyclic Aromatic Hydrocarbons**, or PAHs, are getting renewed attention, with the discovery of more and more of these complex molecules out in the cosmos. Our very own target of the month, the Iris Nebula, is suspected to contain such PAHs, the potential building blocks of life. Kind of makes me wish I had paid more attention in high school chemistry class. Kind of.

Don't forget to keep looking up!

Carl Svensson
Vice-President
Ephemeris Editor

Target of the Month: Iris Nebula

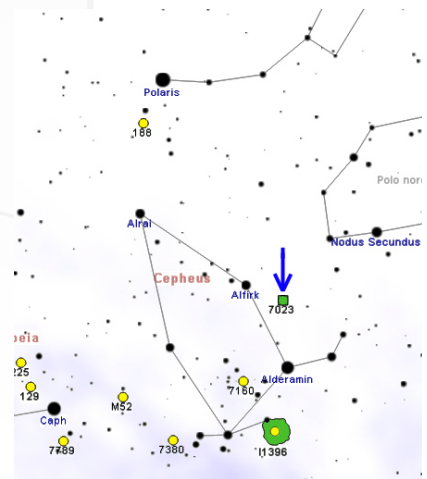
The Iris Nebula, also known as Caldwell 4, NGC 7023, and LBN 487, is a bright reflection nebula in the constellation Cepheus. Situated about 3.5 degrees southwest of Beta Cep (Alfirk), the Iris is illuminated by the relatively young star named V380 Cep. The young star is surrounded by dust that reflects the nearby starlight in characteristic blue tones. Among the nebulosity is a star cluster that adds further interest to the view.

At a magnitude of somewhere between 6.8 and 7.2, the Iris is a difficult visual target in small scopes, but shines gloriously in photographs. A 10" scope should reveal bright visual views of the nebula, and good conditions can bring out the dark patches amidst the nebulosity. In a smart scope, or other long exposure photos, the bright blue color and endlessly intricate dark patches give the imagination a lot of fun shapes to play with.

From our vantage point on Earth, the nebula lies behind clouds of dust, which partly obscure it from view. This same dust also lends its own beauty to the image. These patches of obscuring dust even have their own name. So-called “dark nebulae” are often very rewarding targets in and of themselves. Many of which are catalogued in the Lynd's Catalog of Dark Nebulae (LDN). Lynd also has a catalogue of bright nebulae, LBN, of which the Iris is a proud member.

Top: NGC 7023 by Carl Svensson <https://app.astrobin.com/i/4v6s5o>. Bottom: Map by Roberto Mura, CC BY-SA 3.0, via Wikimedia Commons

Magnitude	7.2
Size	18' x 18'
Object Type	Nebula
Constellation	Cepheus
Distance	1400 ly
Age	5,500 years



Announcements

Swap Meet The semi-annual SJAA Swap Meet returns on September 7th at 9AM in the hallowed halls of the Houge Park Neighborhood Center, Building 1. Join us at our headquarters for a morning of buying, selling, and socializing. Bring your own well-loved, well-functioning gear for potential buyers. The meet runs until noon. If you sleep in, you may miss the swap meet, but you can still make it to Solar Sunday at 2pm. See you there!

New Renewal System Some of you may have noticed that we have changed how new members sign up, and how returning members renew their memberships. This system provides us with better automation and record keeping, which helps us provide robust and timely services for our members. If you have any questions or face any issues, please reach out to us at asksjaa@sjaa.net. As always, you can reach our renewal and sign up system through the website at <https://www.sjaa.net/join-the-sjaa/>.

Discord Exciting news! SJAA is coming to Discord. This popular, community-based platform connects members to SJAA leadership and to each other. Our discord server complements our email offerings, and comes with goodies like observing forecasts, photos of the day, and more. Look for more information soon in your inbox.

We Always Need More Volunteers There are many volunteer opportunities in SJAA! While we love having seasoned experts in our volunteer ranks, we happily welcome enthusiastic beginners, too. Opportunities range from operating telescopes, to authoring content for publication, to helping with event logistics. Please reach out to us at volunteer@sjaa.net to get started!



SJAA Sept 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	1 Venus 1.5° S of the Pleiades  68%	2  77%	3  85%	4 Pluto 0.06° N of Moon  92%	5  97%	6 Board Meeting 6:00PM Guest Speaker 8:00PM  97%
7 Fall Swap Meet 9:00AM Solar 2:00 PM  100% Loaner Exchange, FixIt 2:00PM	8 Neptune 3° S of Moon Saturn 4° S of Moon  100%	9 William Cranch Bond's Birthday (1789)  96%	10  90%	11  82%	12 Moon 1° N of the Pleiades Mars 2° N of Spica  72%	13 Starry Nights Star Party 8:30PM  60%
14 John Dobson's Birthday (1915)  49%	15 James Christy's Birthday (1938)  37%	16 Imaging SIG 7:30PM H.A. Rey's Birthday (1898) Jupiter 5° S of Moon  26%	17 Moon 2° N of M44  17%	18  10%	19 Astro 101 7:00PM ITSP 8:30PM Venus 0.8° S of Moon  4%	20 Member Observing - Mendoza 7:45PM Saturn at opposition  1%
21  0%	22 Autumnal Equinox 11:19AM  1%	23 Spica 1.1° N of the Moon Neptune at opposition  4%	24 Mars 4° N of Moon Geoffrey Burbidge's Birthday (1925)  9%	25  15%	26  23%	27 Astronomy Day Antares 0.6° N of Moon  32%
28  41%	29 Geoffrey Marcy's Birthday (1954)  51%	30 Phil Plait's Birthday (1964)  61%				
			Times and dates are subject to change. Please check sjaa.net and www.meetup.com/sj-astronomy for the latest updates.			
			Meteor Shower Dark Sky Window			